RP4 Tests for Ions – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: Cations and Carbonate **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Result for Mg^{2+} with NaOH(aq)				White precipitate
2	Ionic equation for Mg^{2+} with NaOH(aq)				$Mg^{2+} + 2OH^- \rightarrow Mg(OH)_2$
3	Reagent to test for Ba^{2+}				Dilute H_2SO_4
4	Result for Ba^{2+} with dilute H_2SO_4				White precipitate
5	Test for NH_4^+				Warm with NaOH(aq); damp red litmus turns blue
6	Ionic equation for the NH_4^+ test				$NH_4^+ + OH^- \rightarrow NH_3 + H_2O$
7	Test for OH^-				Red litmus turns blue
8	Test for CO_3^{2-}				Add dilute acid; gas turns limewater cloudy
9	Ionic equation for carbonate with acid				$CO_3^{2-} + 2H^+ \rightarrow CO_2 + H_2O$
10	Equation for limewater turning cloudy				$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

RP4 Tests for Ions – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 2: Sulfate and Halides **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Test for SO_4^{2-}				Acidified $BaCl_2$; white precipitate
2	Ionic equation for the sulfate test				$Ba^{2+} + SO_4^{2-} \rightarrow BaSO_4$
3	Acid used to acidify $BaCl_2$				Dilute HCl
4	Why $BaCl_2$ is acidified				Removes CO_3^{2-} so white $BaCO_3$ cannot form (false positive)
5	Ionic equation for the false positive				$Ba^{2+} + CO_3^{2-} \rightarrow BaCO_3$
6	Test for halide ions				Acidified $AgNO_3$
7	Why $AgNO_3$ is acidified with HNO_3				Removes CO_3^{2-} so Ag_2CO_3 cannot form
8	Ionic equation for the halide test				$Ag^+ + X^- \rightarrow AgX$
9	Order of anion tests				Carbonate, sulfate, halide
10	Why that order is used				Carbonate and sulfate also form precipitates

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.